Smart Autonomous Shopping Cart Project

Logbook

Student Name : Vignesh Lakshmanasamy

Module : PDE4435 – Robotic System Integration

Role : Robot Design & Simulation Environment Development

Project Understanding & Concept Design

24/3/2026

- Reviewed coursework Requirements
- Understood deliverables and expectations
- Spoke with the team members about the cw
- Initiated a git repo

25/3/2026

- As a team – decided to continue with the smart cart simulation
- Prof. asked question to approve the project
    - What is the problem you are going to solve?
        - Handling the cart in supermarket – for elderly, for people shopping with baby etc.
    - What are the existing similar solutions available?
        - Reviewed some IEEE papers, very few in real world as solutions
    - What differentiate your solution to those existing ones?
        - Planning to create completely autonomous – following robot
    - How are you going to build your robot, hardware or simulator (which platform)?
        - Jay and me – Suggest going with hardware development. But unfortunately, all team members are in different location, in different situation. We decided to do in simulation.
    - What kind of guidance do you need to do your project?
        - As it was too early, we didn't think about exact guidance we need.

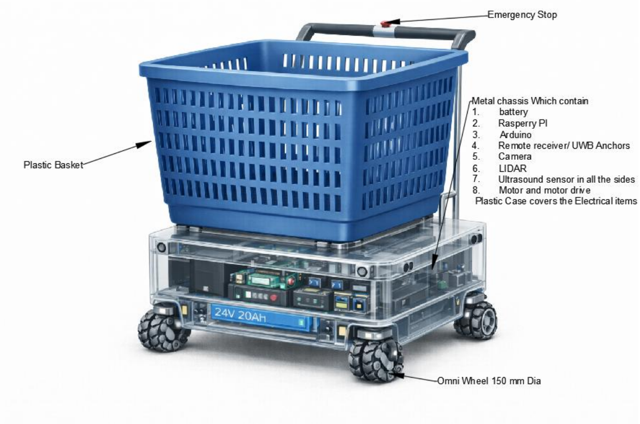
29/3/2026

- Finalized **Smart Autonomous Shopping Cart**
- Identified Modules:
    - Robot Design
    - Navigation
    - Detection
    - Simulation

31/3/2026

- Assigned roles within team
- Decided to go with ROS2 and Gazebo

1/4/2026

- Decided to have the CW 1 design as a reference
- 

- Team members were also ok with that
- I suggested with doing the design in solidworks and convert it to URDF.
- Jay and Ashwin like to use Xacro – as they used it in previous courseworks
- Decided to do have options – 1. In SW@urdf and another one in xacro. Which works fine to keep it
- Answered the Prof. Dr. Judhi – 5 question about the project

Outcome :

- Project Scope Finalized
- Development approach confirmed

Research and Concept Development:

3/4/2026

- Reviewed the grocery, supermarket environment and its constraints
- Went to carrefour – analysed the floor layout and how the people move on the market.
- Saw some baby refuse to sit on the trolley and wants to run or play inside the market.
- Parents are forced to hold baby on one hand and cart on another.

5/4/2026

- Went to lulu, safari and nesto to analyse it floor layout.
- In these supermarkets the space between the shelf is less compared to carrefour.
- More people and carts as it were week off.
- Few people stopped the cart in-between the shelfs to take something. If someone wants to cross that area, have to move the parked cart to pass it.
- Some worst scenario – lady with her 1 year baby came for shopping alone. Baby was in her own baby trolley. She cant take another cart , so she stocked the items in the baby trolley and came for billing

6/4/2026

- Spoke with jay about the sensor that we need to use in the cart
- Analysed their size to fit in the cart
- Lidar, Ultrasound sensor are chosen

7/4/2026

- I suggested to use camera and ml to the team. In an teams call – Ashwin was not ok with that. As it may complicate the progress.
- Shared some links regarding the IEEE paper reference to salaby in whatsapp to add it in the report

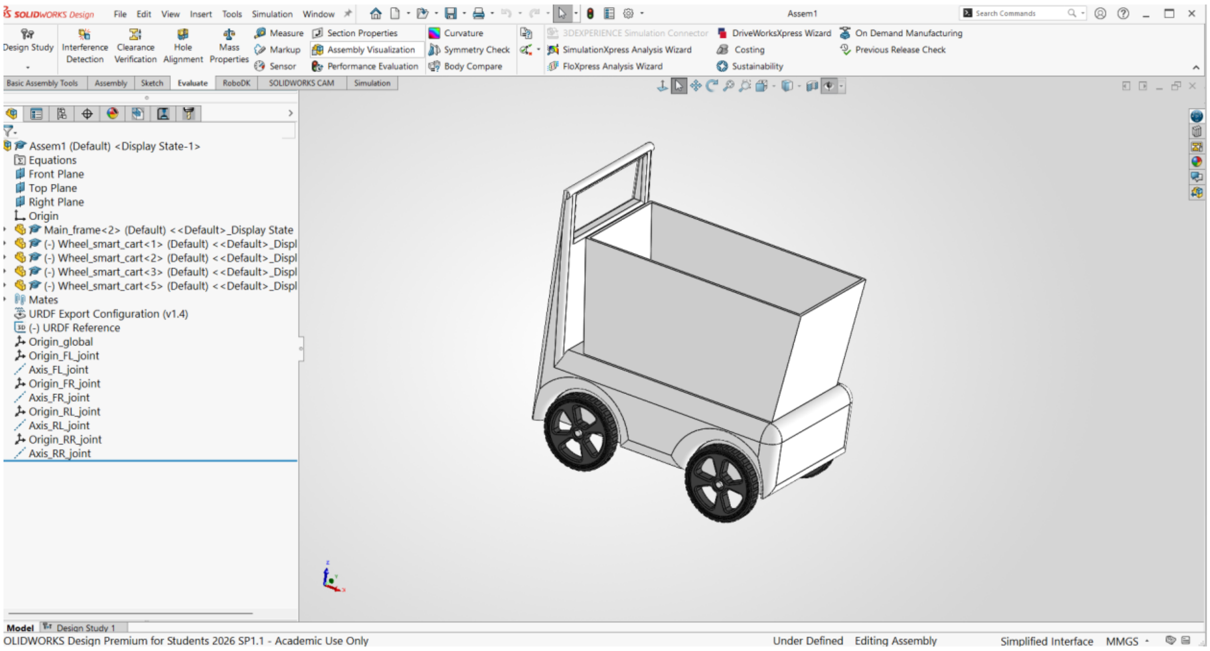
9/4/2026

- Studied about ROS 2 and Gazebo
- Tried to learn the difference between Ros 1 and Ros 2 through some youtube videos

10/4/2026

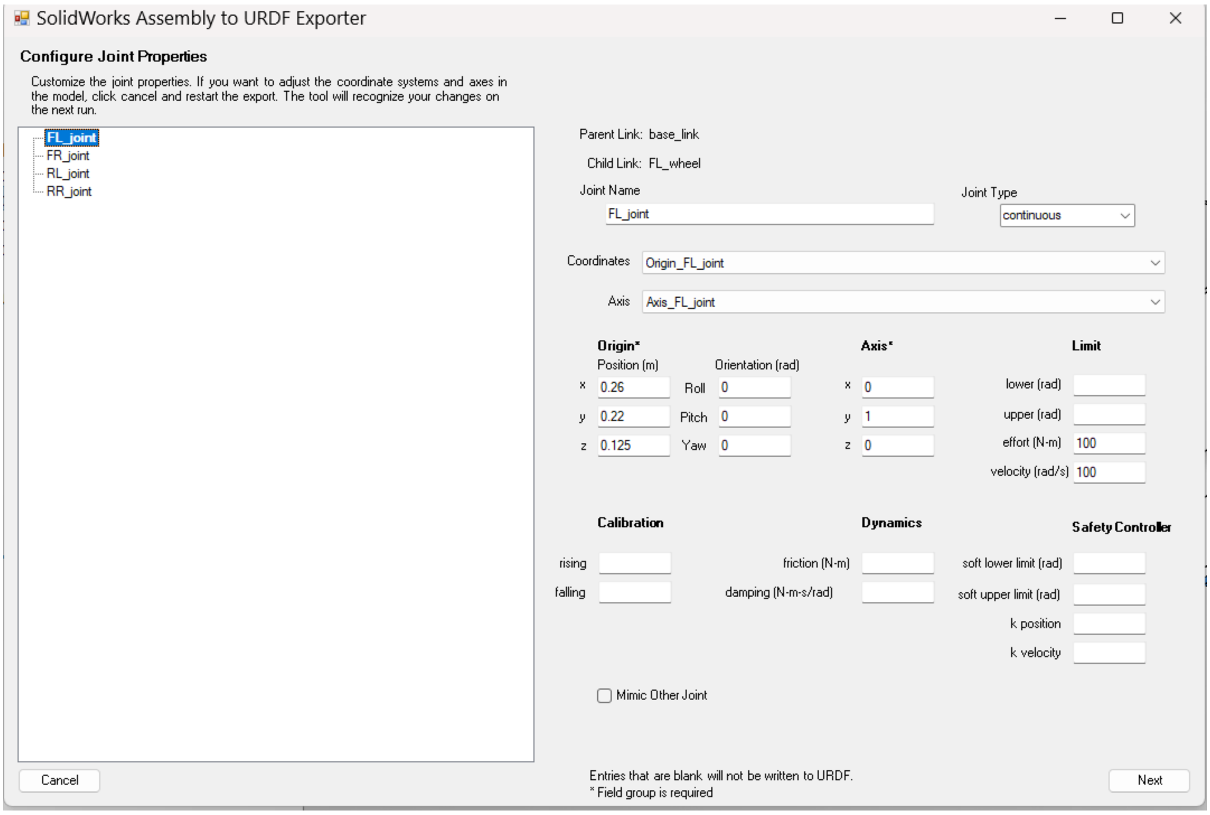
- Installed Ubuntu 24 in VMware
- Installed ROS2 Jazzy in VMware
- Installed all the required packages to start the programming

11/4/2026

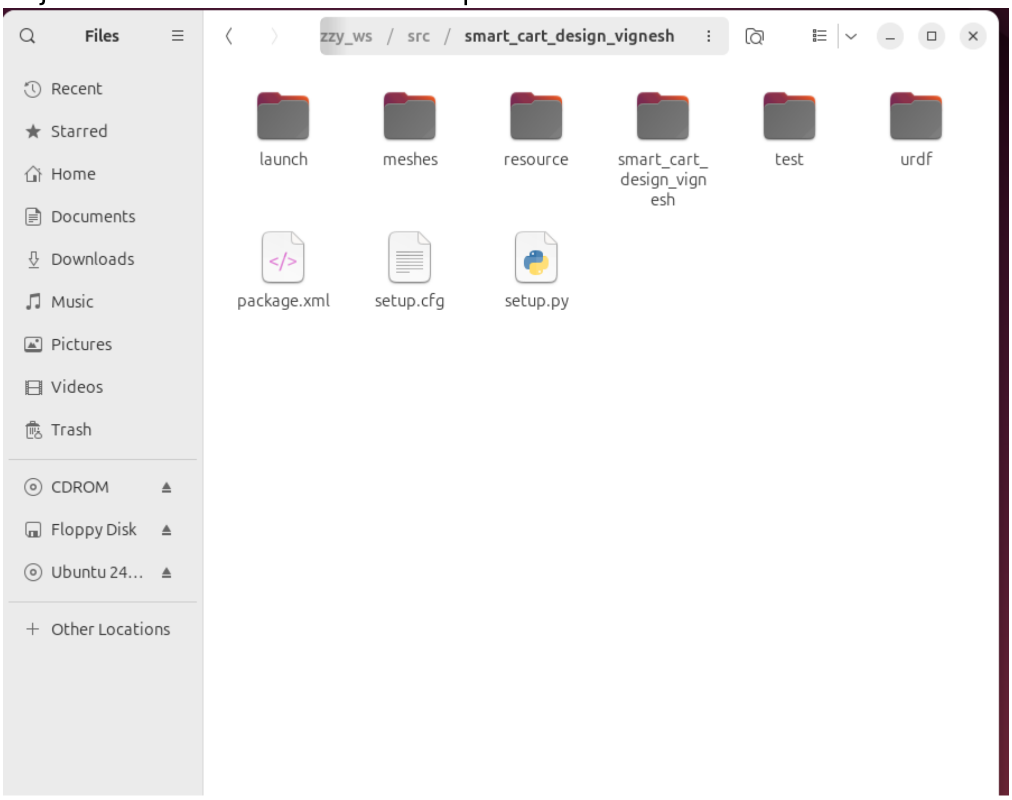
- Created the Cart in Solidworks
- 

12/4/2026

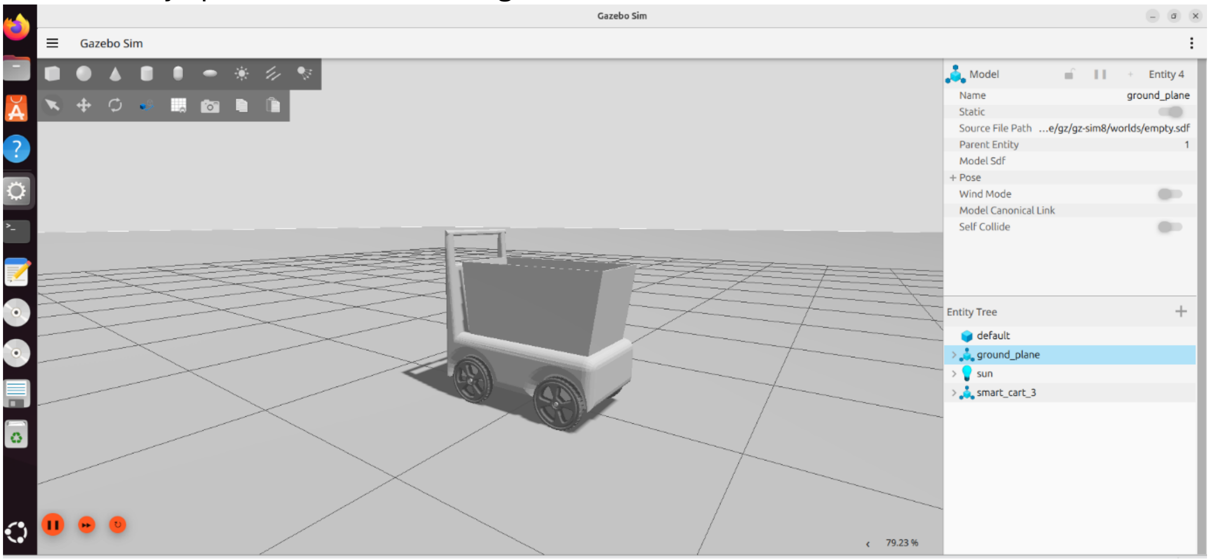
- Used SW@urdf t convert the Solidworks file to URDF

- 

14/4/2026

- Created a workspace in Ros 2
- Took the URDF and mesh files to Ros2
- Adjusted the code to make it compatible to use in ros2
- 

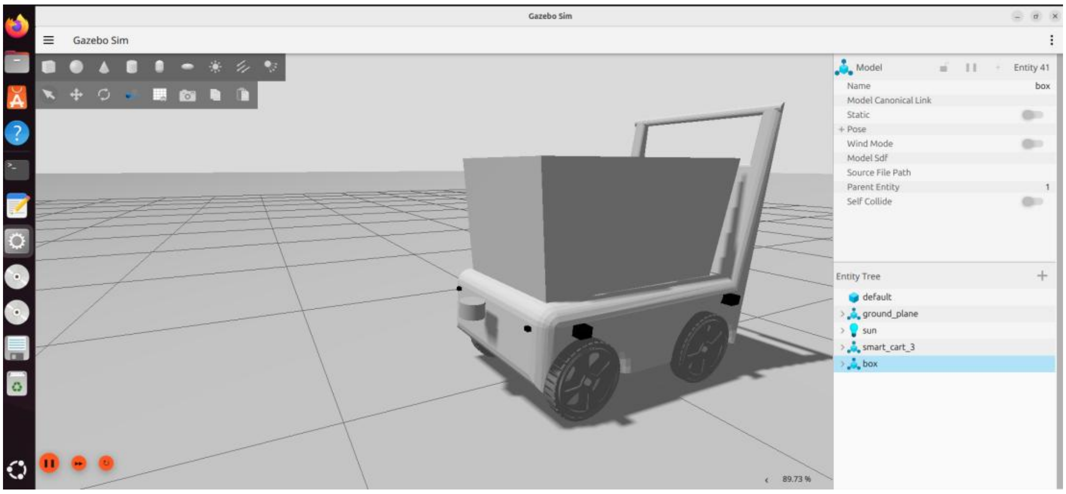
17/4/2026

- Add the Collision Details and other essential codes to the URDF file
- Created a Lauch file for urdf in gazebo
- Successfully spawn the robot in the gazebo
- 

18/4/2026

- Added the Gazebo plugin for Differential drive system.
- Created a new program for the teleop – to test the robot working.
- It didn't work like ros 1
- Here I learned about the bridge function.
- In Ros2 we need a bridge function to transfer the cmd_vel command

19/4/2026

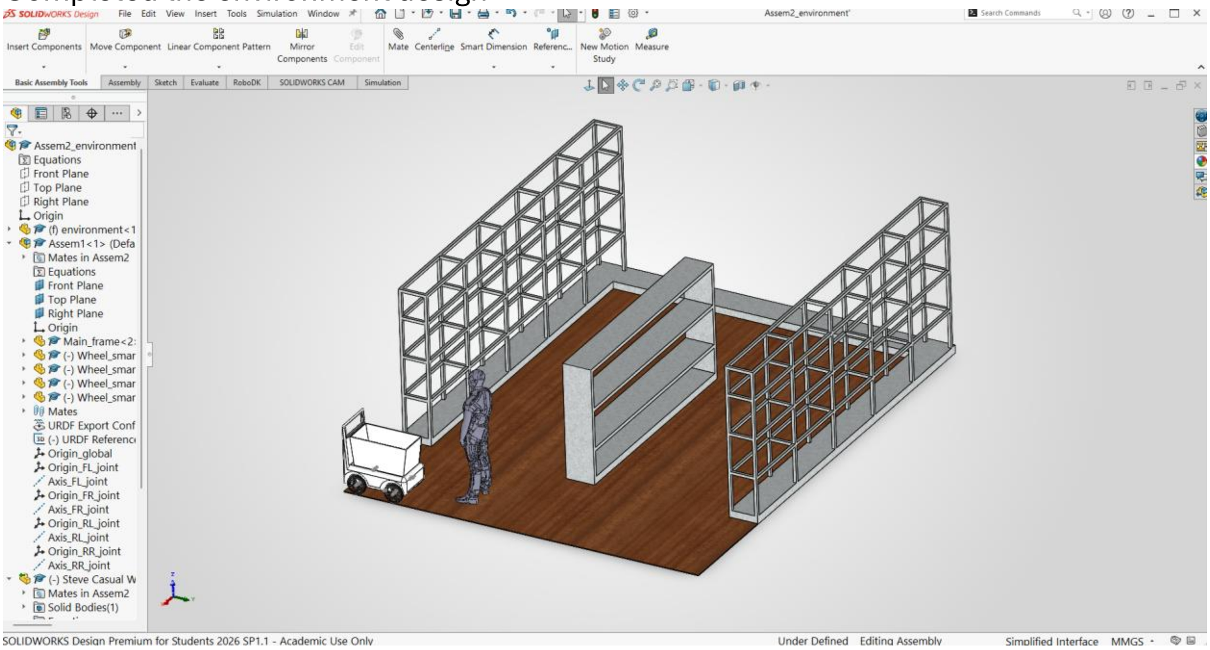
- Added the plugin for the Lidar, ultrasound sensor and UWB sensor
- I discussed with jay for the positioning of the sensor as he was doing the sensor program.
- Jay also created a xacro code for the urdf to have a supporting model
- Both of ur finialised the location of the sensor
- 

21/4/2026

- Send the URDF to Jay – he started integrating the navigation and slam to the urdf.
- I started developing the environment to replicate the supermarket setup.

24/4/2026

- Completed the environment design



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25/4/2026 -28/4/2026

- Working on the logbook
- Planning to keep both xacro and urdf robots in the gazebo, with different launch files
- Facing difficulties with linking other programs
- Creating slides for the presentation
- Reading the report to do the finalizing